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I confirm that I understand my coursework needs to be submitted online via Google Classroom under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a marks of zero will be awarded.

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1.1. Work Breakdown Structure (WBS)

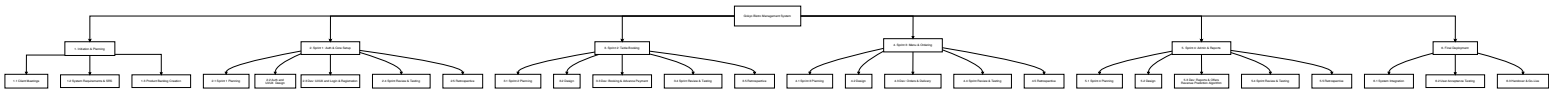

Figure 1: WBS diagram

1.2. Gantt Chart

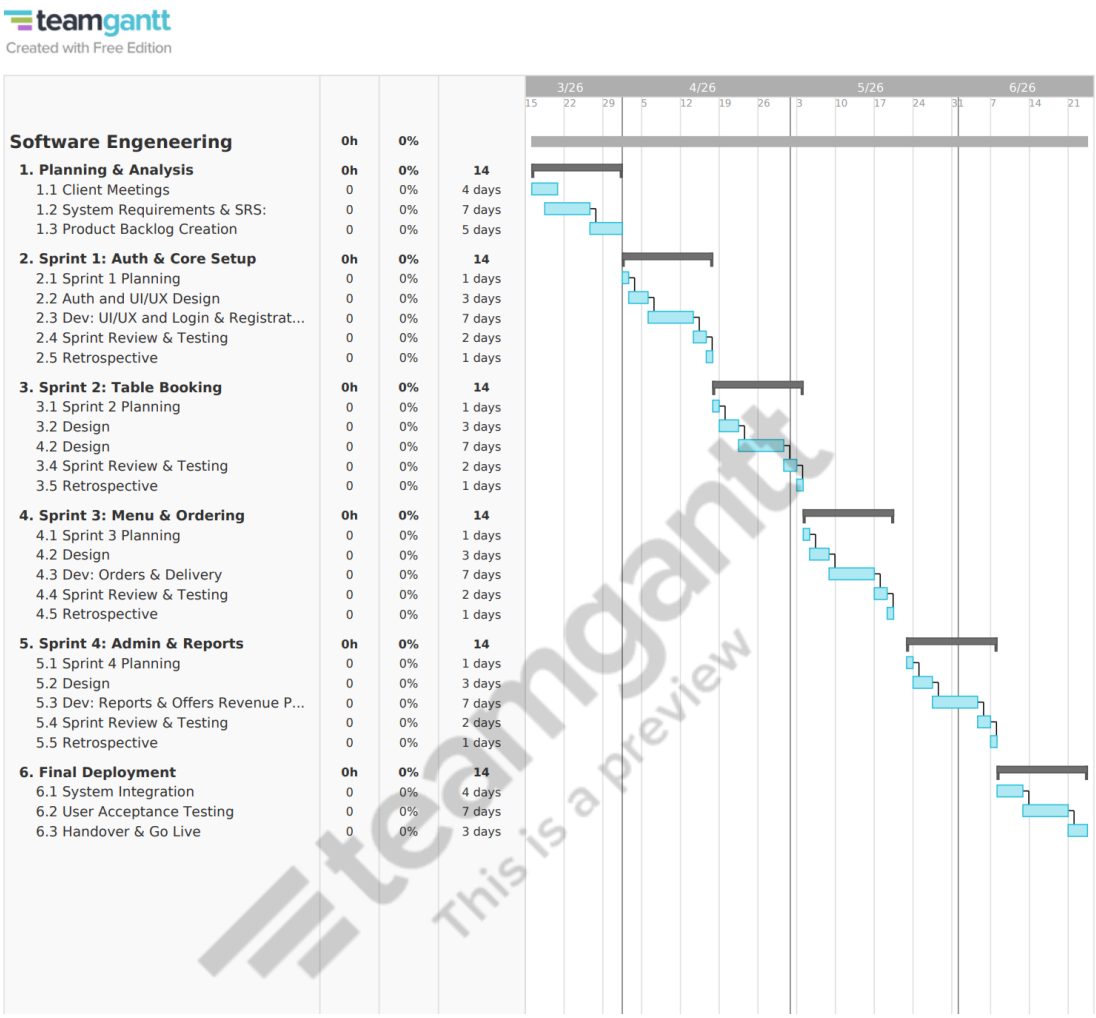

Figure 2: Gantt Chart

2. Use Case Diagram and Descriptions

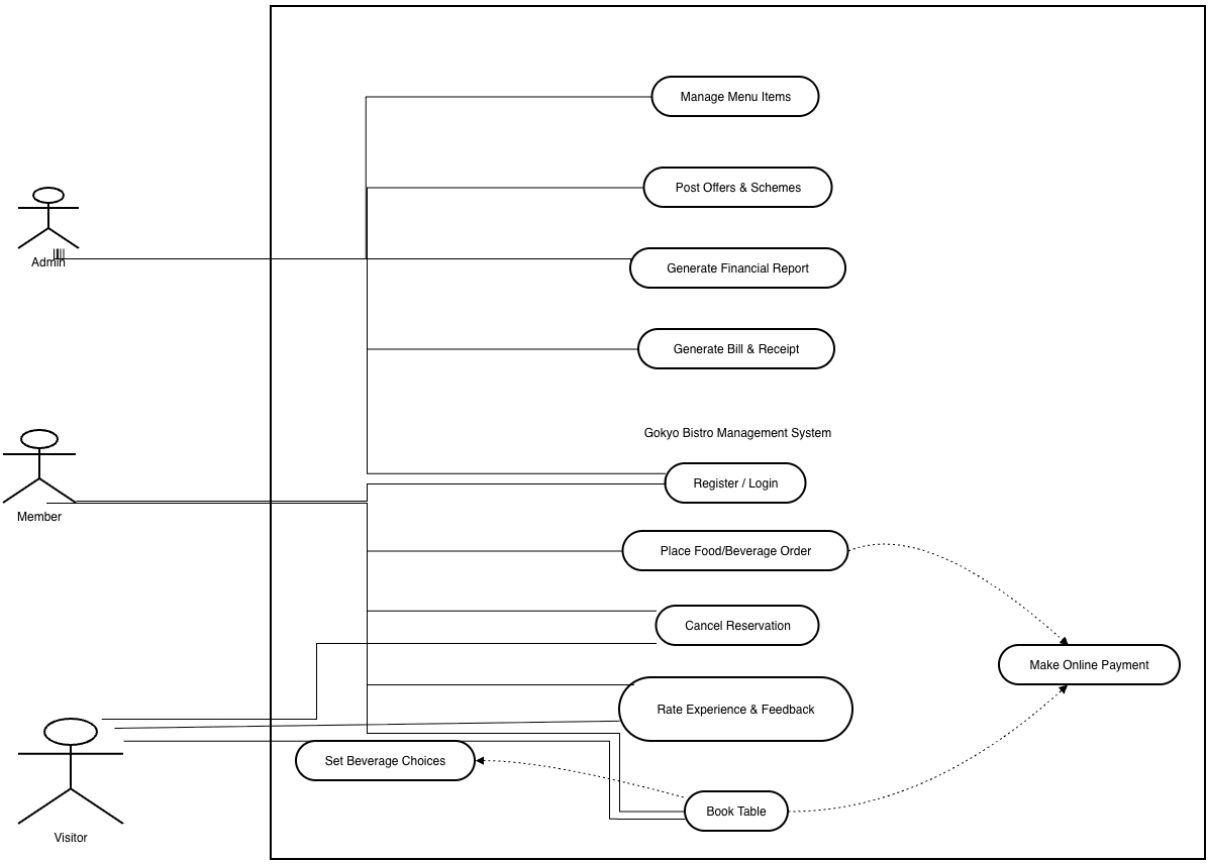

Figure 3: Use Case Diagram

2.1. Use Case: Register / Login

Actors: Admin (initiator), Member (initiator)

Description: An Admin or Member provides their credentials to the system to authenticate and gain access to their respective dashboards.

2.2. Use Case: Manage Menu Items

Actors: Admin (initiator)

Description: An Admin adds, updates, or deletes food and beverage items from the restaurant's active menu.

2.3. Use Case: Post Offers & Schemes

Actors: Admin (initiator)

Description: An Admin inputs and publishes special discount offers and promotional schemes to the system for customers to see.

2.4. Use Case: Generate Financial Report

Actors: Admin (initiator)

Description: An Admin requests a financial report, and the system provides revenue predictions for future years along with menu item recommendations.

2.5. Use Case: Generate Bill & Receipt

Actors: Admin (initiator)

Description: An Admin selects an order, and the system generates an official bill and a receipt for the customer.

2.6. Use Case: Book Table

Actors: Visitor (initiator), Member (initiator)

Description: A customer checks the availability of tables and provides a desired date and time to reserve a table. They pay the advance fee and receive confirmation.

2.7. Use Case: Set Beverage Choices

Actors: Visitor (initiator), Member (initiator)

Description: While booking a table, a customer selects their expected beverage choices from an available list, or reserves specific premium wines/whiskeys if they are a premium member.

2.8. Use Case: Cancel Reservation

Actors: Visitor (initiator), Member (initiator)

Description: A customer selects their active reservation and cancels it. The system processes the cancellation and calculates the appropriate refund based on the cancellation date.

2.9. Use Case: Place Food/Beverage Order

Actors: Member (initiator)

Description: A Member selects items from the menu, chooses a delivery option (dine-in or home delivery), makes an online payment, and receives order confirmation.

2.10. Use Case: Make Online Payment

Actors: Visitor (initiator), Member (initiator)

Description: A customer provides their payment details to pay the mandatory 1200 advance for a table reservation or to pay for a food order.

2.11. Use Case: Rate Experience & Feedback

Actors: Visitor (initiator), Member (initiator)

Description: A customer inputs ratings and written feedback regarding the food, staff behavior, and overall ambience of the restaurant.

3. Expanded Use Case Descriptions

Expanded Use Case 1: Book Table

3.1. Use Case: Book Table

Actors: Visitor (initiator), Member (initiator)

Description: A customer checks the availability of tables and provides a desired date and time to reserve a table. They pay the advance fee and receive confirmation.

Typical Course of Events:

Actor Action	System Response
1. The customer uses the interface to select the "Book Table" option.	2. Presents a calendar and available time slots.
3. Chooses a desired date, time, and party size.	4. Verifies availability and prompts for optional beverage selection.
5. Declines beverage selection and proceeds to confirm booking details.	6. Indicates the mandatory 1200 advance fee and requests payment details.
7. Pays the 1200 advance fee via the payment gateway.	8. Provides confirmation of the reservation and generates a booking ID.

Alternative Courses:

- **Line 4:** The customer chooses to pre-select beverages. The system presents the beverage menu. The customer selects items (or premium wines if a premium member). The use case resumes at step 6.
- **Line 4:** There are no tables available for the selected date and time. The system displays an "Unavailable" message and suggests alternative slots. Use case ends or restarts at step 3.
- **Line 7:** The customer's payment fails or is rejected. The system displays an error message. Use case ends.

Expanded Use Case 2: Place Food/Beverage Order

3.2. Use Case: Place Food/Beverage Order

Actors: Member (initiator)

Description: A Member selects items from the menu, chooses a delivery option (dine-in or home delivery), makes an online payment, and receives order confirmation.

Typical Course of Events:

Actor Action	System Response
1. The Member clicks on the menu and adds desired food/beverage items to their cart.	2. Presents the cart summary and asks for the order type (Dine-in or Home Delivery).
3. Chooses "Home Delivery" and inputs their delivery address.	4. Calculates the total bill and requests payment details.
5. Pays the total bill via the online payment gateway.	6. Provides an order confirmation receipt and routes the order details to the kitchen.

4. Sequence Diagram

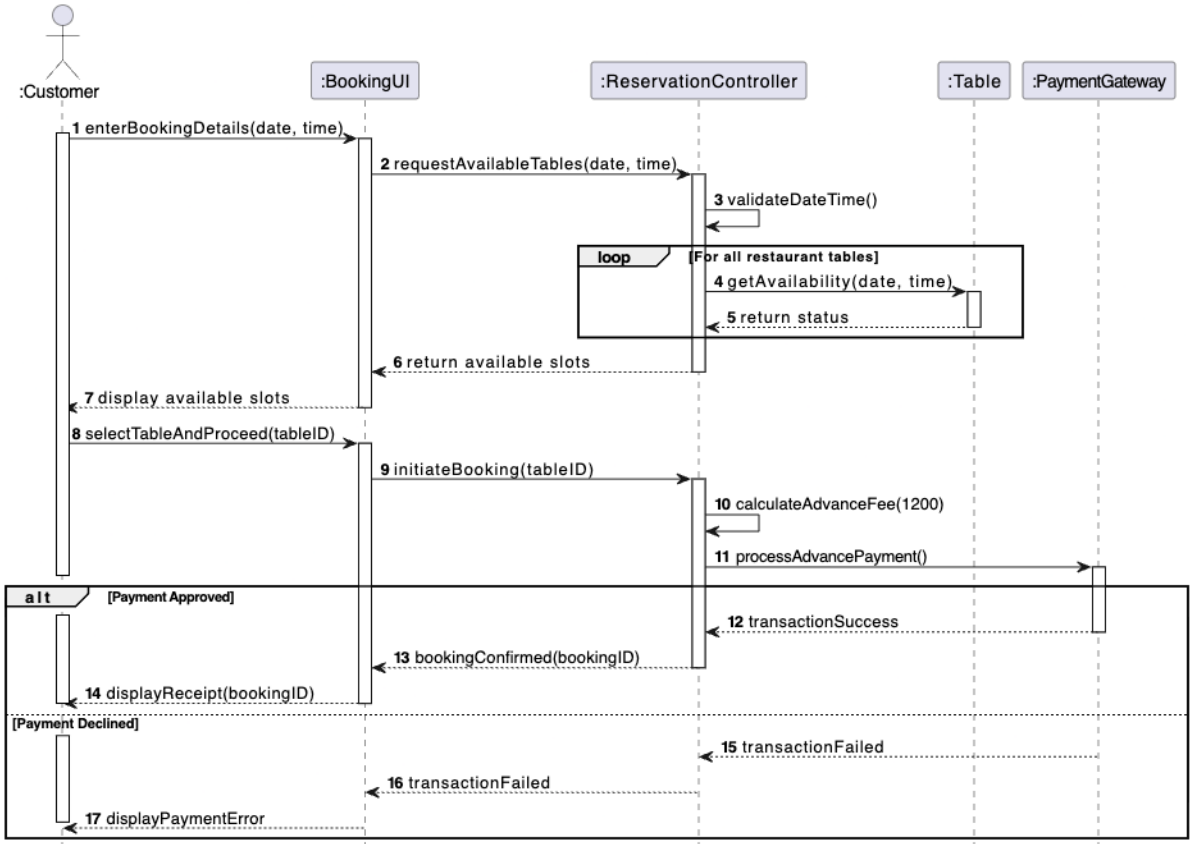

Figure 4: Sequence diagram

5. Activity Diagram

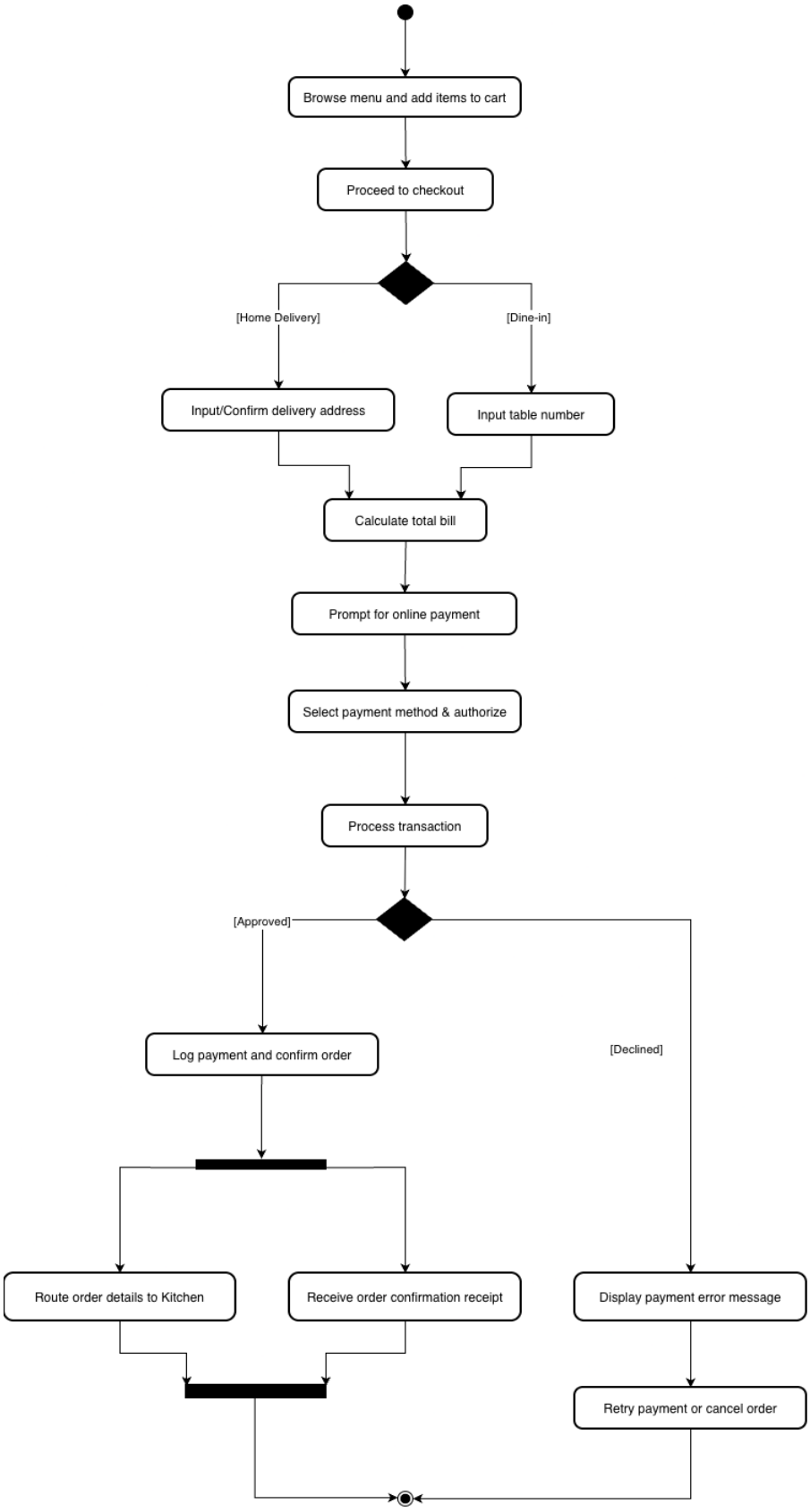

Figure 5: Activity diagram

6. Analysis Class Diagram

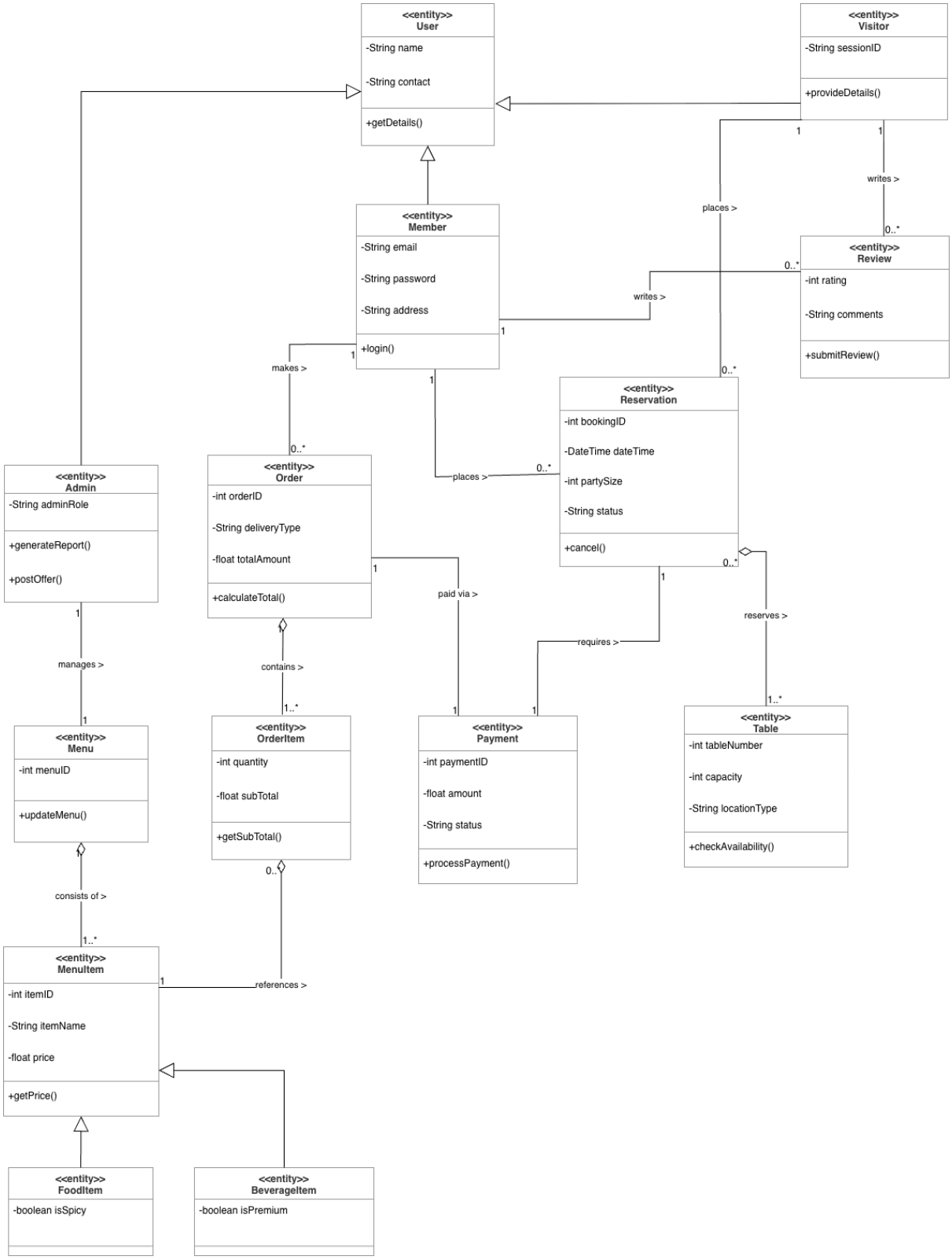

Figure 6: Analysis Class Diagram

7. UI/UX Prototyping

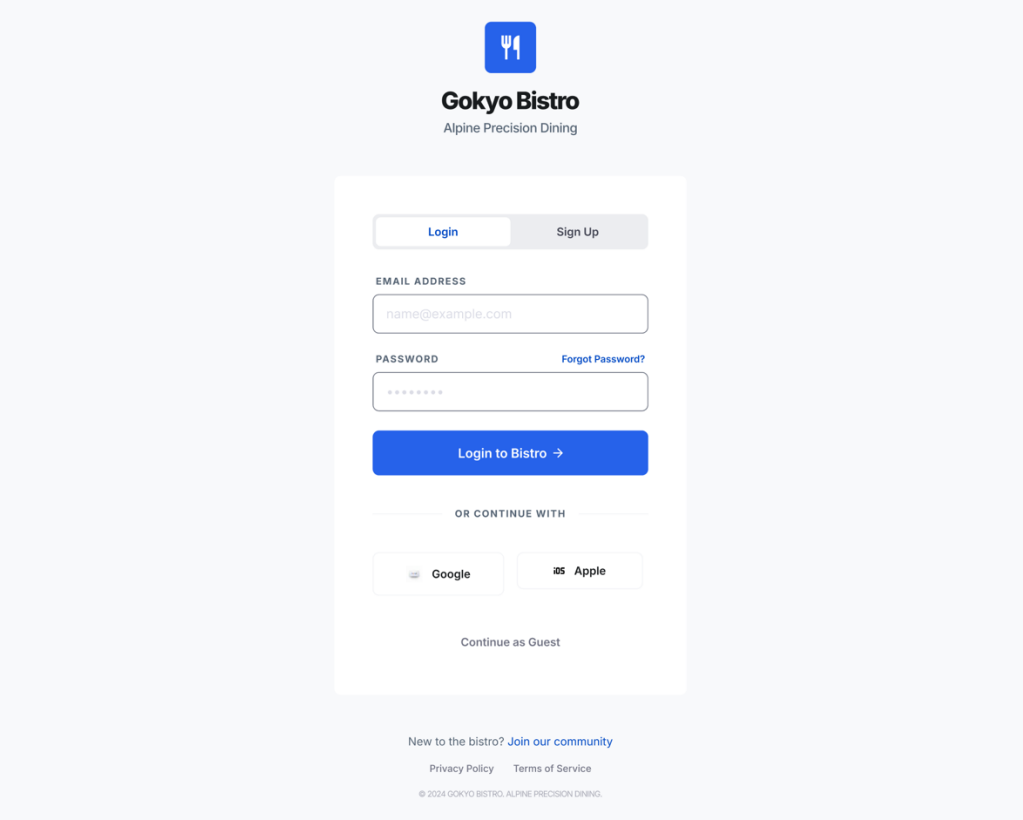

Figure 7: Login Page

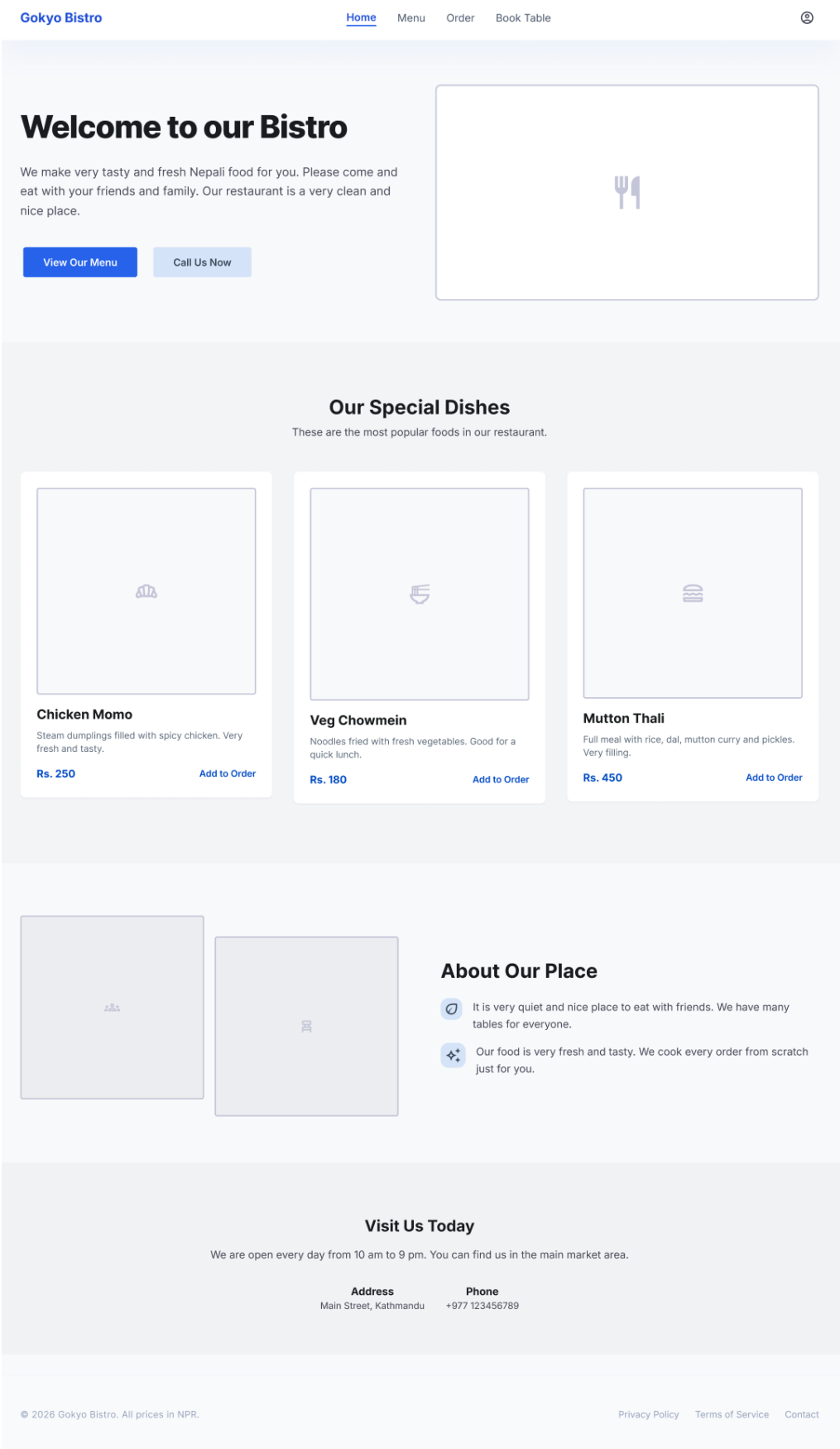

Figure 8: Home Screen

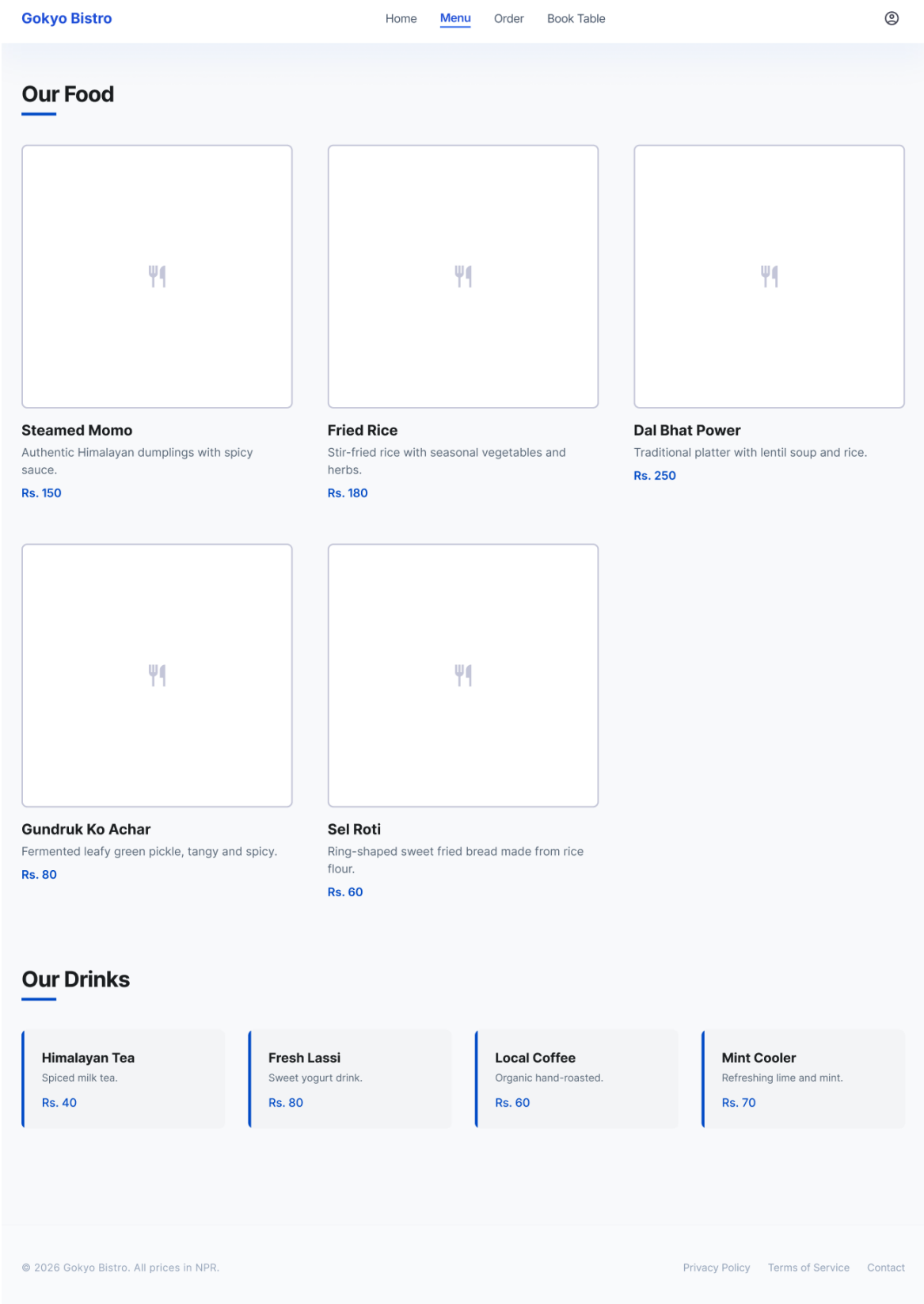

Figure 9: Menu page

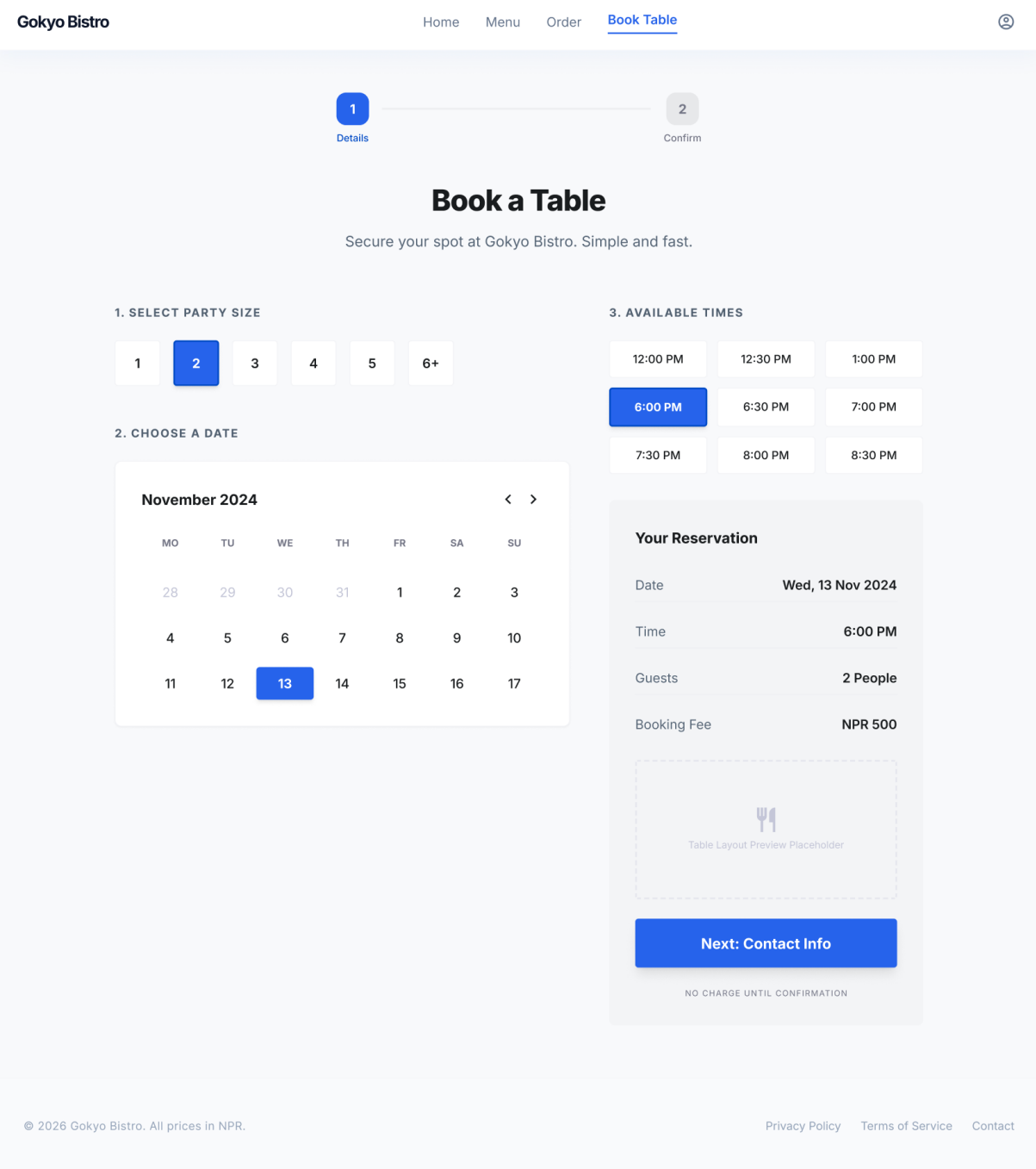

Figure 10: Table Booking

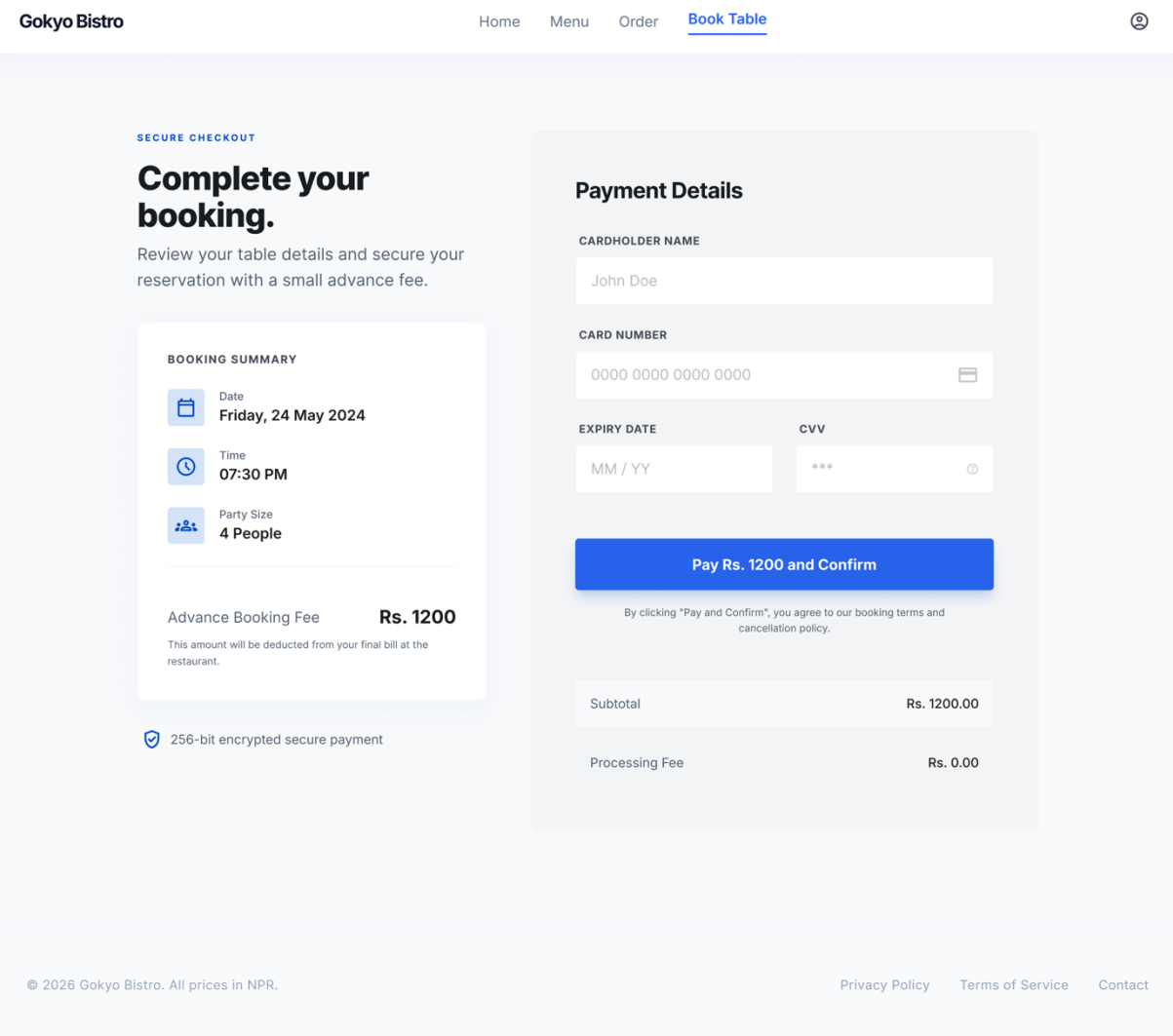

Figure 11: Complete payment Details

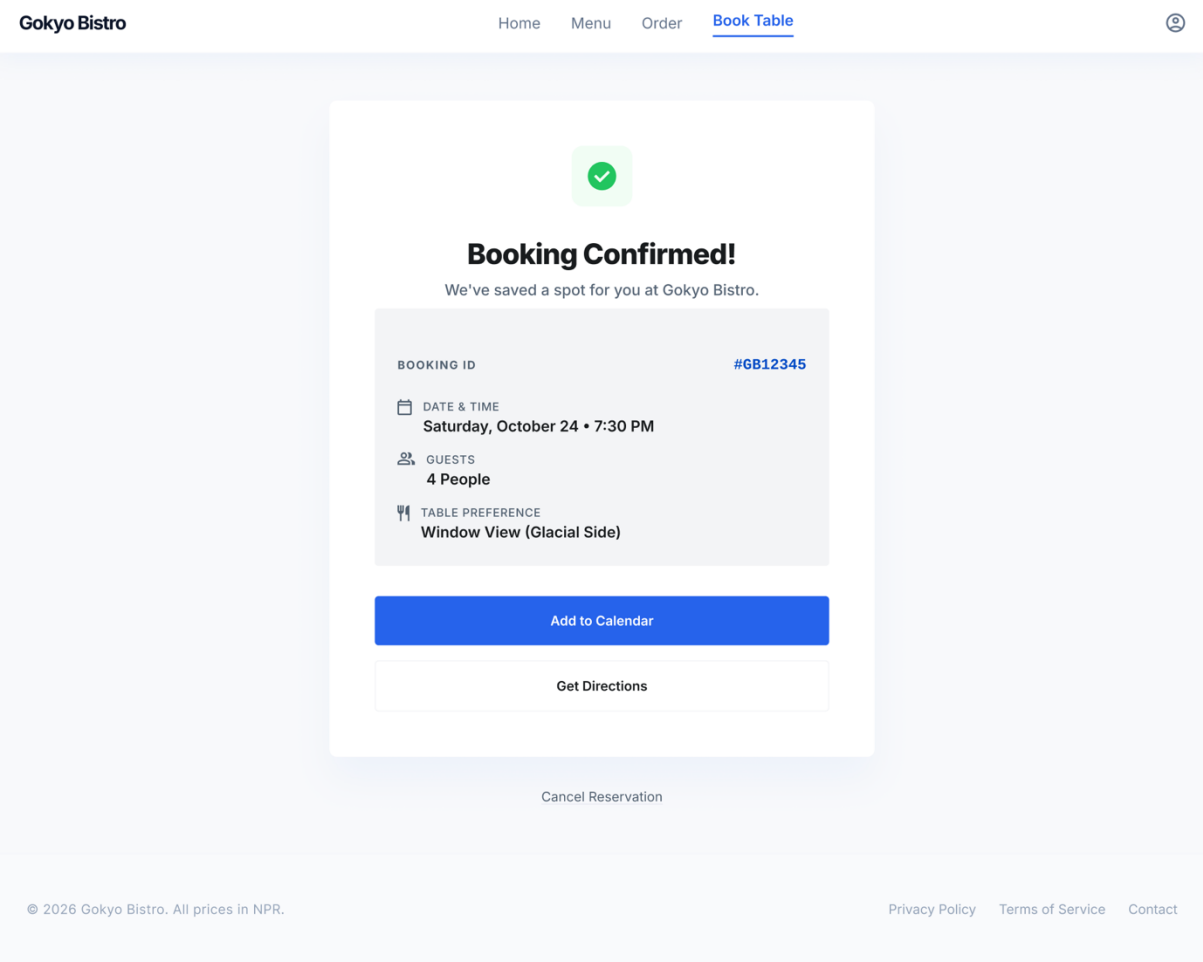

Figure 12: Booking Confirmed

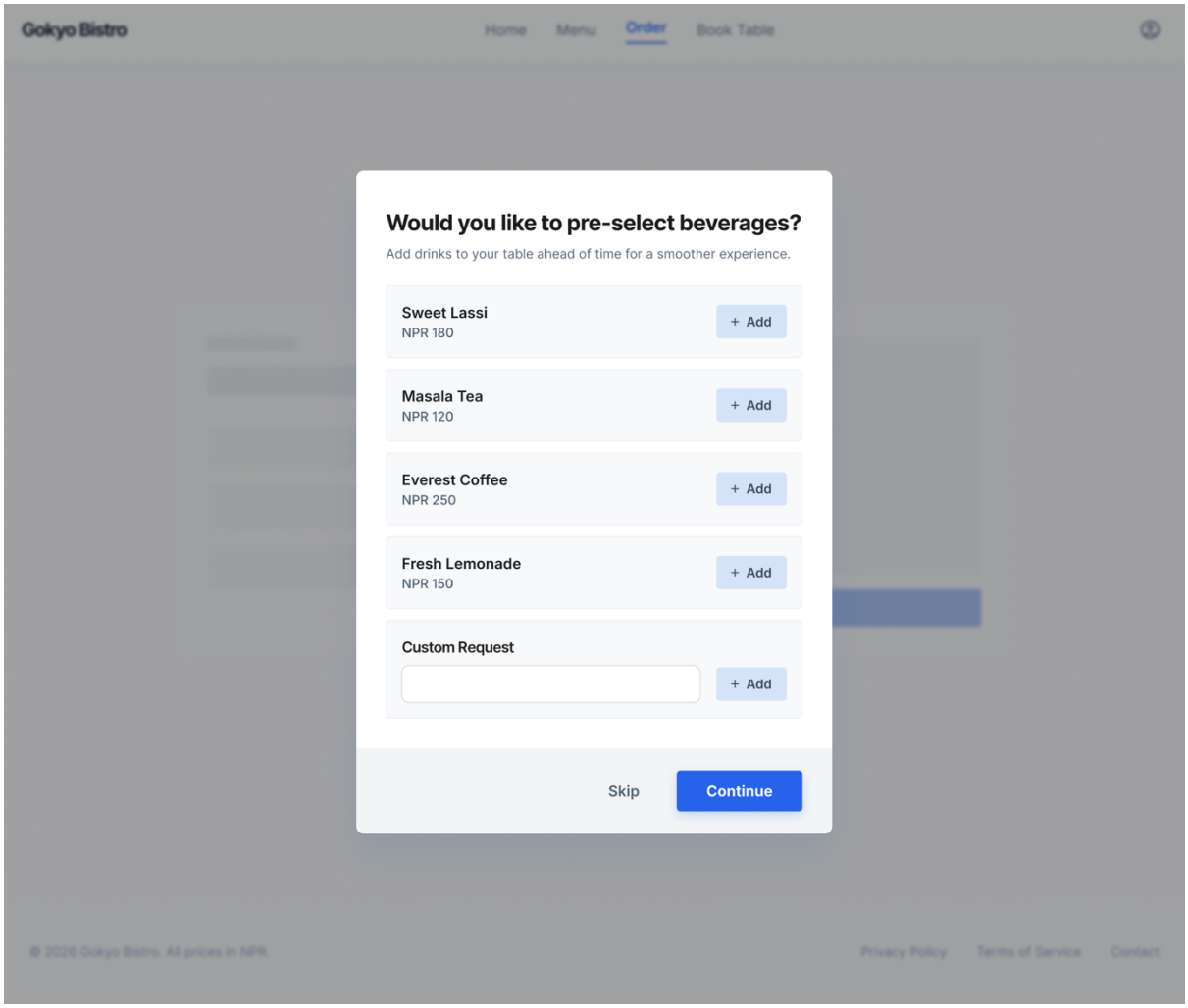

Figure 13: Optional Beverage Choices